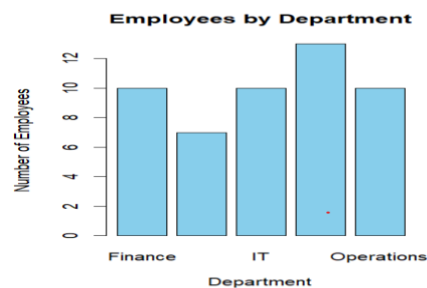


TYPES OF VISUALISATIONS FOR EMPLOYEE SALARY DATASET

1. Bar Chart (Department)

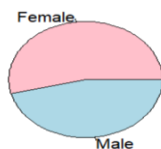
```
barplot(table(employee$Department),  
        col="skyblue",  
        main="Employees by Department",  
        xlab="Department",  
        ylab="Number of Employees")
```



2. Pie Chart (Gender)

```
pie(table(employee$Gender),  
    col=c("pink","lightblue"),  
    main="Gender Distribution")
```

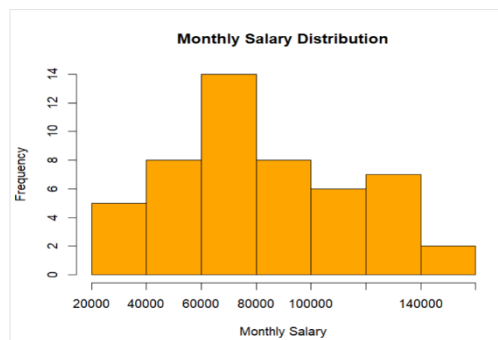
Gender Distribution



3. Histogram (Monthly Salary)

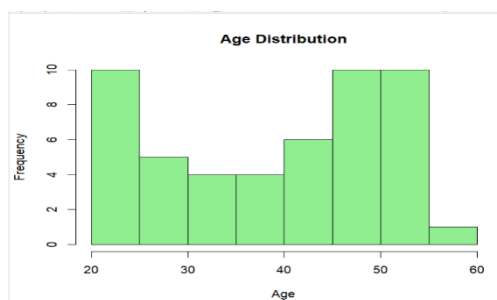
```
hist(employee$Monthly_Salary,  
      col="orange",  
      main="Monthly Salary Distribution",  
      xlab="Monthly Salary",
```

```
border="black")
```



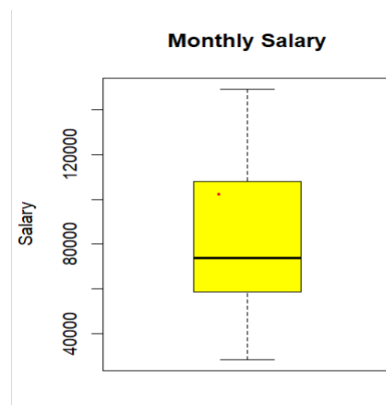
4. Histogram (Age)

```
hist(employee$Age,  
      col="lightgreen",  
      main="Age Distribution",  
      xlab="Age",  
      border="black")
```



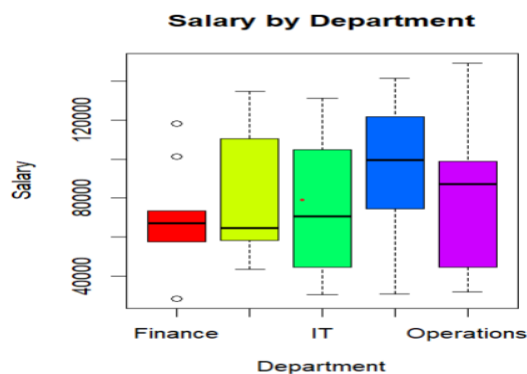
5. Box Plot (Salary)

```
boxplot(employee$Monthly_Salary,  
         col="yellow",  
         main="Monthly Salary",  
         ylab="Salary")
```



6. Box Plot (Salary by Department)

```
boxplot(Monthly_Salary ~ Department,  
        data=employee,  
        col=rainbow(length(unique(employee$Department))),  
        main="Salary by Department",  
        xlab="Department",  
        ylab="Salary")
```



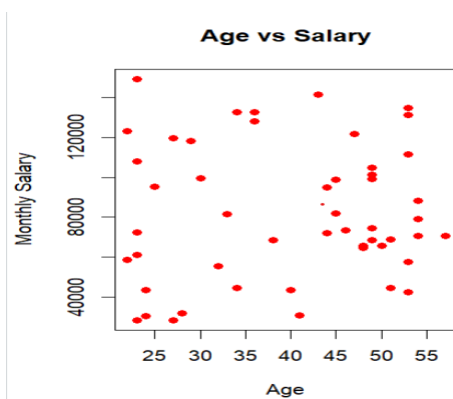
7. Scatter Plot (Experience vs Salary)

```
plot(employee$Experience_Years,  
      employee$Monthly_Salary,  
      pch=19,  
      col="blue",  
      main="Experience vs Salary",  
      xlab="Experience (Years)",  
      ylab="Monthly Salary")
```



8. Scatter Plot (Age vs Salary)

```
plot(employee$Age,  
      employee$Monthly_Salary,  
      pch=19,  
      col="red",  
      main="Age vs Salary",  
      xlab="Age",  
      ylab="Monthly Salary")
```



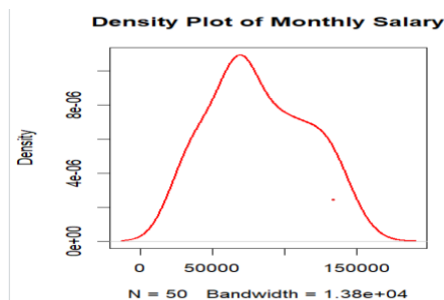
9. Line Chart

```
employee <- employee[order(employee$EmployeeID), ]  
plot(employee$EmployeeID,  
      employee$Monthly_Salary,  
      type="l",  
      lwd=2,  
      col="blue",  
      main="Salary Trend",  
      xlab="Employee ID",  
      ylab="Monthly Salary")
```



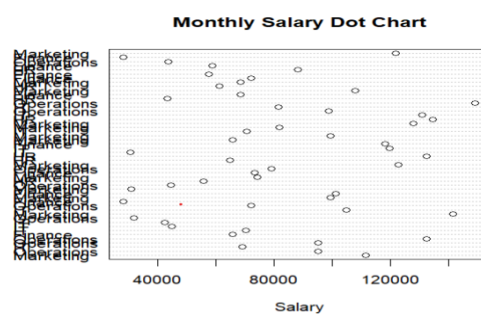
10. Density Plot

```
plot(density(employee$Monthly_Salary),  
     col="red",  
     lwd=2,  
     main="Density Plot of Monthly Salary")
```



11. Dot Chart

```
dotchart(employee$Monthly_Salary,  
          labels=employee$Department,  
          main="Monthly Salary Dot Chart",  
          xlab="Salary")
```



12. Horizontal Bar Chart (City)

```
barplot(table(employee$City),  
        horiz=TRUE,  
        col="cyan",  
        main="Employees by City",
```

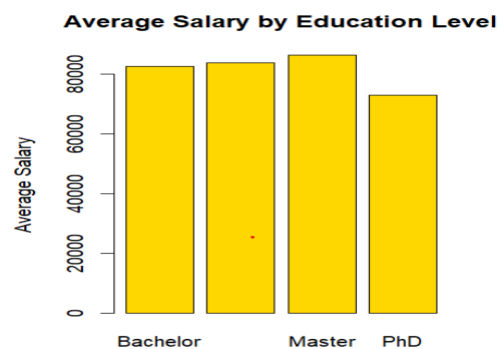
```
xlab="Number of Employees")
```



13. Average Salary by Education Level

```
avg_salary <- aggregate(Monthly_Salary ~ Education_Level,  
  data=employee,  
  mean)
```

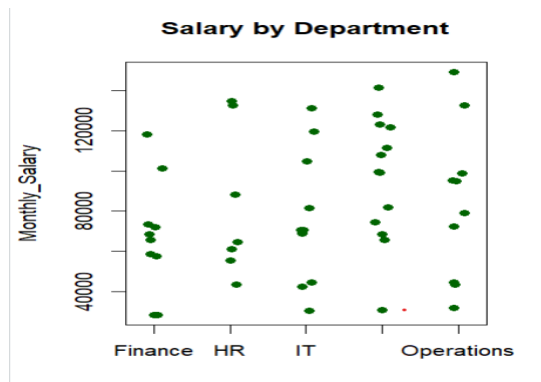
```
barplot(avg_salary$Monthly_Salary,  
  names.arg=avg_salary$Education_Level,  
  col="gold",  
  main="Average Salary by Education Level",  
  ylab="Average Salary")
```



14. Strip Chart

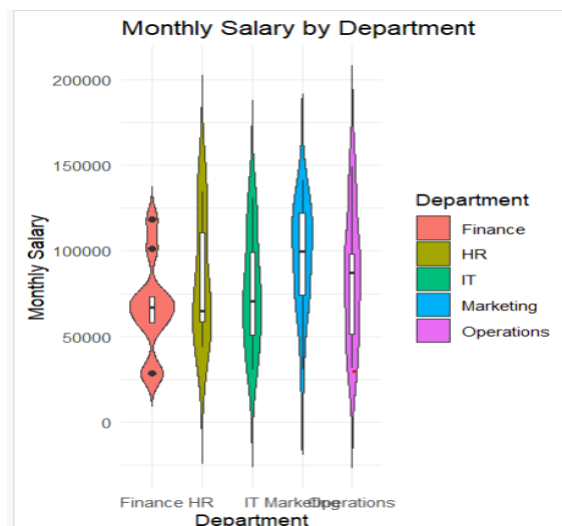
```
stripchart(Monthly_Salary ~ Department,  
  data=employee,  
  method="jitter",  
  vertical=TRUE,  
  pch=19,  
  col="darkgreen",
```

```
main="Salary by Department")
```



15. Violin Plot

```
ggplot(employee,
  aes(x=Department,
    y=Monthly_Salary,
    fill=Department))+
geom_violin(trim=FALSE)+
geom_boxplot(width=0.1, fill="white")+
labs(title="Monthly Salary by Department",
  x="Department",
  y="Monthly Salary")+
theme_minimal()
```

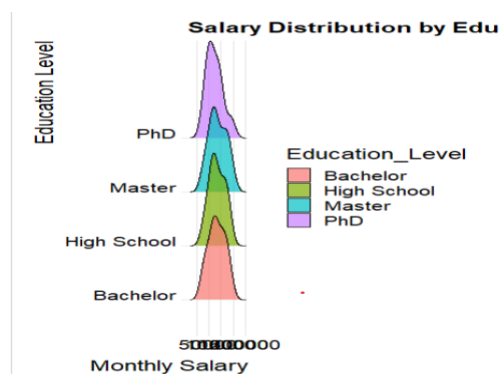


16. Ridgeline Plot

```
library(ggplot2)
```

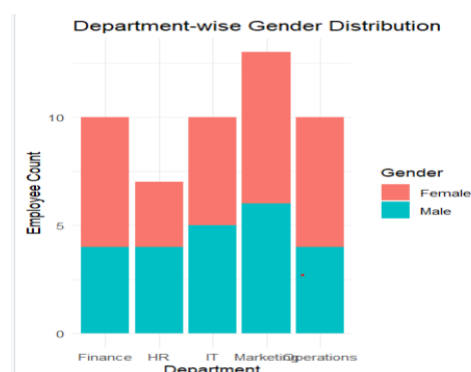
```
library(ggbridges)
```

```
ggplot(employee,
  aes(x = Monthly_Salary,
      y = Education_Level,
      fill = Education_Level)) +
  geom_density_ridges(alpha = 0.7, scale = 1) +
  labs(title = "Salary Distribution by Education Level",
      x = "Monthly Salary",
      y = "Education Level") +
  theme_ridges()
```



17. Stacked Bar Chart

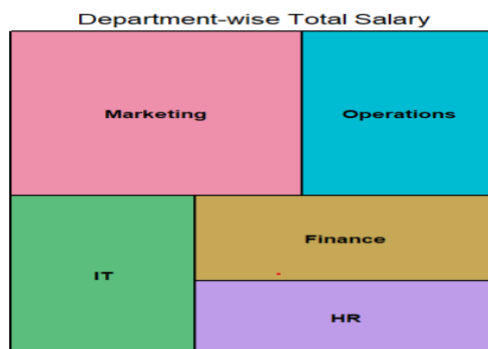
```
ggplot(employee,
  aes(x=Department,
      fill=Gender))+
  geom_bar()+
  labs(title="Department-wise Gender Distribution",
      x="Department",
      y="Employee Count")+
  theme_minimal()
```



18.Treemap

```
library(treemap)

treemap(
  dept_salary,
  index = "Department",
  vSize = "Monthly_Salary",
  title = "Department-wise Total Salary"
)
```



19.Bubble Plot

```
ggplot(employee,
  aes(x=Experience_Years,
    y=Monthly_Salary,
    size=Age,
    color=Department))+
  geom_point(alpha=0.7)+
  labs(title="Experience vs Salary",
    x="Experience (Years)",
    y="Monthly Salary")+
  theme_minimal()
```

